Qi Sun

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Research Interests: Efficient and reliable language-model agents, with a focus on on-policy distillation, model merging, and retrieval for clinical applications.

EDUCATION

New York University Shanghai Shanghai, China
B.S. in Computer Science Expected May 2027
Independent capstone: *Right Finding, Wrong Organ: Auditing Anatomical Scope in Clinical Report Retrieval*

PUBLICATIONS & PREPRINTS

[1] **Qi Sun**[*], Siyue Zhang[*†], Yulin Chen, Yuxiang Xue, Ru Peng, Chen Zhao[†]. "From 'Weak' Signals to Strong Models: Preference Delta Aggregation with LoRA Merging." **NeurIPS 2026.** arXiv:2606.00357.

[2] Yizhou Liu[*], **Qi Sun**[*], Yulin Chen, Siyue Zhang, Chen Zhao. "Search, Do not Guess: Teaching Small Language Models to Be Effective Search Agents." *Committed to EACL 2027 (decision pending)*. arXiv:2604.04651.

[3] **Qi Sun**[*], Siyue Zhang[*†], Tingyu Song, Yilun Zhao, Chen Zhao[†]. "Rethinking Offline On-Policy Distillation: From Scalar Advantages to Distributional Supervision." *Under review at ACL Rolling Review (October 2026)*.

[*]Equal contribution. [†]Corresponding author.

RESEARCH EXPERIENCE

Undergraduate Researcher 2024.06 – Present
Advisor: Prof. Chen Zhao NYU Shanghai

Distributional on-policy distillation for agentic search (DistOPD) [3] 2026.06 – Present

- Found that offline on-policy distillation (Lightning OPD) improves math reasoning but consistently falls below its SFT starting point in multi-turn agentic search, and traced the failure, empirically and theoretically, to distributional flattening under scalar-advantage supervision on fixed rollouts.
- Developed DistOPD, an offline method that keeps teacher and reference probabilities over a small set of candidate tokens plus a residual token, so the teacher scores each rollout once and is never queried during training.
- Matched online OPD on agentic search with 10.6× fewer training GPU-hours. Led the project end to end, from failure analysis and method design to experiments and writing.

Preference Delta Aggregation for weak-to-strong training (PDA) [1] 2026.01 – 2026.05

- Proposed PDA, which turns each pair of a weak and a weaker model (e.g., Qwen3-4B over 1.7B) into a preference delta trained as a LoRA adapter, and aggregates many deltas by LoRA merging, with Geometric Alignment Merging (GAM) to reduce interference between adapters.
- Improved Qwen3-8B by 6.8 points on knowledge reasoning and 7.3 points on agentic search (benchmark averages), 2.1 and 4.3 points above the best single-delta baseline. Presenting author at NeurIPS 2026.

Small language models as effective search agents [2] 2025.06 – 2025.12

- Led the on-policy distillation and adaptive search parts of the project. Distilled agentic search from Qwen3-32B into Qwen3-1.7B with on-policy distillation, and designed probing and rejection fine-tuning experiments showing that adaptive when-to-search strategies often hurt small models.
- The paper finds that distilled small models search less and hallucinate more. Training them to search consistently and ground answers in retrieved evidence improves over agent distillation by 17.3 points on Bamboogle and 15.3 on HotpotQA.
- Supported by the Dean's Undergraduate Summer Research Fund.

Research Assistant 2026.05 – Present
Advisor: Prof. Yiqiu (Artie) Shen NYU Langone Health, New York, NY

Right Finding, Wrong Organ: Auditing Anatomical Scope in Clinical Report Retrieval

- Built a feature-oriented retrieval system over abdominal CT/MRI reports and structured EHR that surfaces similar cases for ambiguous pancreatic presentations, using a clinical codebook that encodes findings, quantitative thresholds, and negation for hybrid BM25 and dense retrieval.
- Audited whether retrieved evidence comes from the right anatomical scope, a failure in which a retriever matches the right finding in the wrong organ. This work is also my independent capstone. Manuscript in preparation for a medical informatics journal.

HONORS & AWARDS

- Dean's Undergraduate Summer Research Fund, NYU Shanghai 2025
- ICPC Regional Contest (Ningxia), Bronze Prize 2025
- CSAW LLM CTF 2024 (MENA), Second Place 2024

LEADERSHIP

- **Chair,** NYU Shanghai ACM Student Chapter 2024.06 – 2025.06
- **Director of Competition,** NYU Shanghai ICPC Training Community 2024.06 – 2025.06
  With Prof. Xianbin Gu, built the team's training platform (Vue3, Django) with an online judge and LLM-based personalized problem recommendation.

TECHNICAL SKILLS

- **Languages:** Python, C/C++, Java, SQL, JavaScript
- **ML Stack:** PyTorch, Hugging Face Transformers, PEFT/LoRA, vLLM, FAISS, Slurm multi-GPU clusters